# Enlargements

Exam code: 0607

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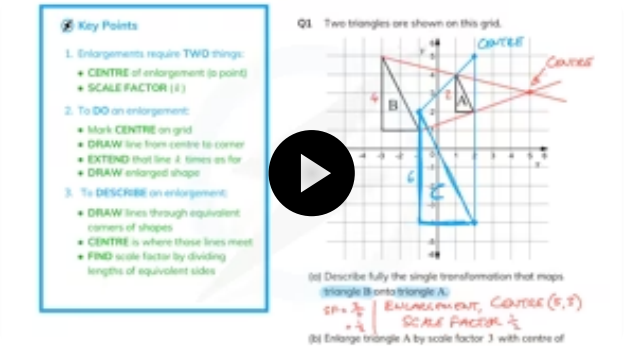
Updated on 10 March 2026

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# Enlargements

## What is an enlargement?

- An **enlargement changes** the **size** and **position** of a shape
- The **length of each side** of the shape is **multiplied** by a **scale factor**
  - If the scale factor is **greater than 1** then the **enlarged image** will be **bigger** than the **original object**
  - If the scale factor is **less than 1** then the **enlarged image** will be **smaller** than the **original object**

- The **centre of enlargement** determines the **position** of the enlarged image
  - If the scale factor is **greater than 1** then the **enlarged image** will be **further away** from the centre of enlargement
  - If the scale factor is **less than 1** then the **enlarged image** will be **closer to** the centre of enlargement

## How do I enlarge a shape?

- STEP 1
  Pick a **vertex** of the shape and count the **horizontal and vertical distances** from the **centre of enlargement**
- STEP 2
  Multiply **both** the horizontal and vertical distances by the given **scale factor**

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- STEP 4
  **Repeat** the steps for the **other vertices**
  - You might be able to draw the enlarged shape from the first vertex by **multiplying the original lengths** by the scale factor
    - This can be done **quickly** if the shape is made up of **vertical and horizontal lines**

- STEP 5
  **Connect** the vertices on the **enlarged image** and label it

## Examiner Tips and Tricks

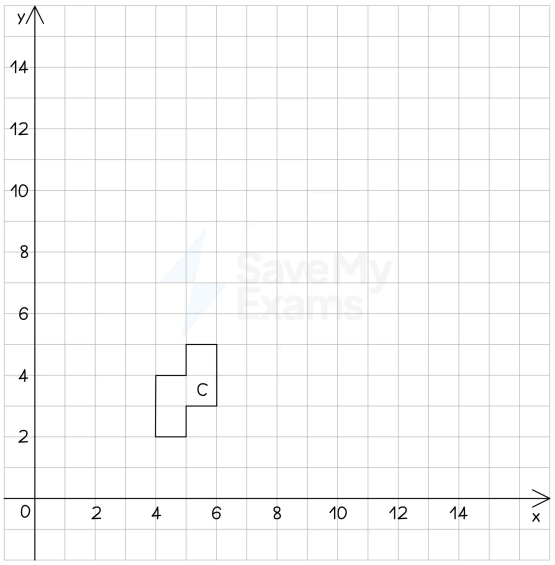
To **check** that you have enlarged a shape **correctly**:

- **Draw lines** going **from the centre of enlargement** to **each of the vertices** of the original shape
- **Extend** these lines
- The **lines should go through** the **corresponding vertices** of the enlarged image

## Worked Example

(a) On the grid below enlarge shape C using scale factor 2 and centre of enlargement (2, 1).

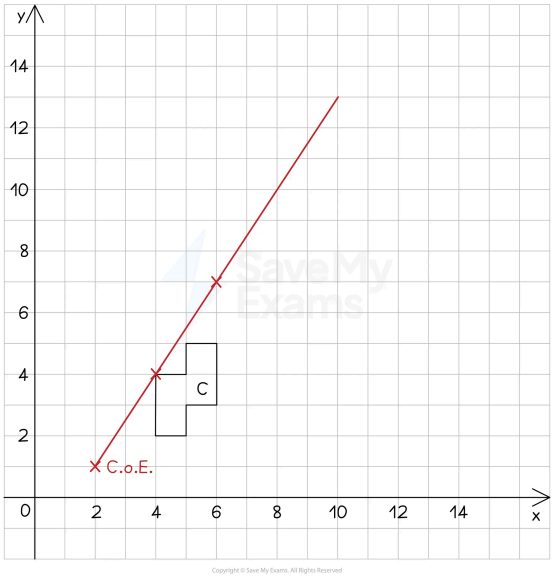
Label your enlarged shape C'.




**Answer:**

Start by marking on the centre of enlargement (CoE).
Count the number of squares in both a horizontal and vertical direction to go from the CoE to one of the vertices on the

Count these new distances from the CoE to the corresponding point on the enlarged image and mark it on.
Draw a line through the CoE and the pair of corresponding points, they should line up in a straight line.




Repeat this process for each of the vertices on the original object (or at least 2).
Join adjacent vertices on the enlarged image as you go.
Label the enlarged image C'.

## How do I describe an enlargement?

- To describe an **enlargement**, you must:
  - State that the transformation is an **enlargement**
  - State the **scale factor**
  - Give the coordinates of the **centre of enlargement**
- To find the **scale factor**:

    - **Divide** the **length of the enlarged side** by the **length of the original side**
- To find the **centre of enlargement**:
    - **Pick a vertex** of the **original shape**
    - Identify the **corresponding vertex** on the **enlarged image**
    - Draw a **line going through these two vertices**
    - **Repeat** this for the **other vertices** of the original shape
    - These lines will **intersect** at the centre of **enlargement**

## How do I reverse an enlargement?

- If a shape has been **enlarged**, you can perform a single transformation to **return the shape** to its **original size and position**
- An enlargement can be **reversed** by multiplying the enlarged shape by the **reciprocal of the original scale factor**
    - The centre of enlargement is the same

- For a shape enlarged by a scale factor of 3 with centre of enlargement (-1, 6)
    - The **reverse transformation** is
        - an enlargement of scale factor $\frac{1}{3}$
        - with centre of enlargement (-1, 6)

### Worked Example

Describe fully the single transformation that creates shape B from shape A.

**Answer:**

We can see that the image is larger than the original object, therefore it must be an **enlargement.**
As the enlarged image is bigger than the original object, the scale factor must be greater than 1.
Compare two corresponding edges on the object and the image to find the scale factor.

The height of the original "H" is 3 squares
The height of the enlarged "H" is 9 squares

$$\therefore \text{Scale Factor} = \frac{9}{3} = 3$$

Draw a straight line through the CoE and a pair of corresponding points on the original object and the enlarged image.
Repeat this step for as many vertices as you feel you need to so you can confidently locate the CoE.
Do this for all pairs of vertices to be sure!
The point of intersection of the lines is the CoE.

**Shape A has been enlarged using a scale factor of 3 and a centre of enlargement (9, 7) to create shape B**

# Fractional enlargements

## How do I enlarge a shape with a fractional scale factor?

- You follow the **same steps** as for other enlargements
- The **only differences** will be the **enlarged shape's size and position**
  - The enlarged shape will be **smaller** than the original
    - These are still called enlargements even though the shape gets smaller
  - The enlarged shape will be **closer to the centre of enlargement** than the original

### Worked Example

On the grid below enlarge shape C using scale factor $\frac{1}{2}$ and centre of enlargement (4, 2).

Write down the four vertices of your enlarged shape.

**Answer:**

Mark the centre of enlargement at (4, 2)
Count the number of squares horizontally and vertically to any vertex - we've chosen the vertex at (-2, 4)

Multiply these distance by the scale factor, $\frac{1}{2}$

$$6 \text{ 'right'} \times \frac{1}{2} = 3 \text{ 'right'}$$

$$2 \text{ 'down'} \times \frac{1}{2} = 1 \text{ 'down'}$$

Count these new distances (which should be smaller than the originals) from the CoE to find the corresponding point on the new image and mark it on

Repeat as required and draw lines through corresponding vertices and the CoE as a check

**The four vertices of the enlarged shape are (0, 0), (0, 4), (1, 3) and (1, 1)**

## How do I describe an enlargement with a fractional scale factor?

- To describe an enlargement, must:
  - State that the **transformation** is an **enlargement**
  - State the **scale factor**
    - This will be a **fraction**
  - Give the coordinates of the **centre of enlargement**
- To find the **scale factor**:
  - **Pick a side** of the **original shape**
  - Identify the **corresponding side** on the **enlarged image**
    - This should be **smaller**
  - **Divide** the **length of the enlarged side** by the **length of the original side**

## How do I reverse a fractional enlargement?

- If a shape has undergone a **fractional enlargement**, you can perform a single transformation to **return the shape** to its **original size and position**
- An enlargement can be **reversed** by multiplying the enlarged shape by the **reciprocal of the original scale factor**
  - the centre of enlargement is the same
- For a shape enlarged by a scale factor of $\frac{1}{2}$ with centre of enlargement (0, 5)
  - The **reverse transformation** is
    - an enlargement of scale factor 2
    - with centre of enlargement (0, 5)

### Examiner Tips and Tricks

- **Multiplying** by the fraction $\frac{1}{n}$ is the same as **dividing** by $n$
  - $6 \times \frac{1}{2}$ is the same as $6 \div 2$
  - This can make the calculations quicker on the **non-calculator paper**

### Worked Example

Describe fully the single transformation that creates shape B from shape A.

**Answer:**

We can see the image is smaller than the original so it is a **fractional enlargement**
Compare two corresponding edges to find the scale factor - we've used the top edge

$$\text{scale factor} = \frac{\text{new edge}}{\text{old edge}} = \frac{2}{6} = \frac{1}{3}$$

Draw straight lines through corresponding vertices on the original shape
Repeat this 3–4 times and you should find the lines intersect at the same point
This point will be the CoE

**Shape A has been enlarged using a scale factor of $\frac{1}{3}$ and a centre of enlargement (3, −3.5) to create shape B**

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